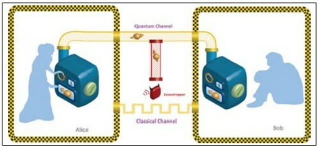

Figure 3: Quantum channel

Alice(Sender)									
Bit	0	1	1	0	1	1	0	0	1
Basis	⊗	○	⊗	⊗	⊗	○	○	⊗	○
Polarity	↑	↖	↓	↑	→	↑	↑	↗	↙
Bob(Receiver)									
Basis	⊗	⊗	○	⊗	⊗	○	○	⊗	○
Polarity	↑	↖	↙	↑	↗	↑	↑	↖	↙
Common Bits	0	Err	Err	0	Err	1	0	Err	1

Figure 4: Table showing qubit matching

for data transmission. The process starts with Alice choosing two strings, X and Y, and then encoding them with qubits. Let us now define a term called *basis*, which is a vector that defines a coordinate system; mathematically, this is a set of linearly independent vectors over a real or a complex plane. Hence, if a vector V $\{v_1, v_2, \ldots\ldots v_n\}$ is finite and X $\{x_1, x_2, \ldots.. x_n\}$ denotes coordinates of vector X, then according to this principle, V forms the *basis* of X if the following two conditions hold:

- If $x_1v_1 + x_2v_2 + x_3v_3 + \ldots\ldots\ldots + x_nv_n = 0$ then $x_1 = x_2 = x_3 = \ldots\ldots\ldots = x_n = 0$
- For all 'x' in X, $x = x_1v_1 + x_2v_2 + x_3v_3 + \ldots\ldots\ldots + x_nv_n$ where xi is called the coordinate of vector X with respect to *basis* V.

In our key distribution, the ith bit of Y decides the *basis* of the ith bit of X. Since each bit of X is encoded using the *basis* of Y, it is practically impossible to decode X without knowing Y. Now, Alice transfers X to Bob over the quantum channel. During this transfer, an eavesdropper, say Eve, might try to obtain a state of X. At this point, three states of X co-exist, one each with Alice, Bob and Eve, while only Alice knows the *basis* Y. Both Bob and Eve predict their own version of Y, say B and E. Using their respective *basis*, Bob and Eve begin generating their own version (or state) of X. Now, Bob broadcasts to Alice, acknowledging his version of X calculated using *basis* B, say X'. Alice and Bob now begin checking or comparing each bit of X and X'. The bits where X and X' are not equal are discarded. Of all the *n* bits in X, let *m* bits match with X'. There is always a lower limit on matching bits, below which the current vector X and its *basis* Y are discarded, and Alice repeats the whole process again. Alice chooses a random number of bits from among the matched bits (usually *m/2*) and declares it as a shared key. During this process of bit matching and declaration, Alice and Bob use another property of analytical physics called privacy amplification, which measures differences in amplitude of the signals transferred from Alice to Bob and vice versa. Any difference in amplitude means suspicious behaviour (eavesdropper) in the channel, and hence a hand break is performed and the whole process is started again. If, at any instant, Eve using her own *basis* E, tries to replicate bits that are acknowledged between Alice and Bob, these bits change their spin, polarity or both at that very instant due to the *no-cloning* theorem; hence, Alice performs a hand break with Bob and repeats the whole process by selecting a new X and Y, and then retransmits. Figure 4 shows a hypothetical example of qubit matching.

Hence, the agreed key between Alice and Bob in the above example is: 00101. As seen in the table in Figure 4, this protocol has single orientation and polarity for a bit. Therefore, the single photon source is used. Practically, a single photon is difficult to emerge, as photons exist in packets (quanta). Several other algorithms have been developed on the basis of BB84, which exploit the collective pattern and behaviour of photons and are under constant improvement. Some algorithms take advantage of the inherent co-relation of the polarity of photons (entanglement); such algorithms make a guess of bit orientation at Bob's end, knowing its orientation at Alice's end. Again, in such cases, the probability of eavesdropping or counterfeiting is nullified by the *no-cloning* property of the photons.

Similar to the process of key distribution, data encoded with qubits can be transferred. One major drawback in data transfer is the error rate. Even the smallest disturbance in orientation will lead to a complete retransfer of data, creating bottlenecks in the quantum channel. Various error correcting codes are being researched but are still far from being implementable or even discussed in detail.

Limitations of quantum cryptography

1. Computers capable of transferring quantum information over a quantum channel are very large, complex and costly. Hence, only major IT firms, networking giants and a few well-supported educational institutes can pursue R&D in quantum cryptography. This leads to a monopoly and non-standard quantum information theories.
2. The error rate increases exponentially even if a fraction of sunlight interferes with the optic fibre cable. Even the most advanced shielding that exists today does not guarantee zero interference.
3. With increasing distances, qubits tend to become more error-prone. Amplifiers cannot be used in a quantum network, as it would make eavesdropper detection a difficult task. As a result, after a few hundred miles, the error rate becomes so high that reconstruction of qubits using entanglement is practically impossible. END

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